Exercise Sheet 21

In Exercises 21.1–21.4 and 21.6, $G = \mathrm{SL}(2,\mathbb{R})$, $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$, $K = \mathrm{SO}(2)$, and $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$. The matrices are defined in Exercises 21.1 and 21.2. Later exercises refer back to them.

Exercise 21.1. Write $c = \cos\theta$ and $s = \sin\theta$. Let

$$k_\theta = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and $E = \tfrac{1}{2}(H + iS)$, $F = \tfrac{1}{2}(H - iS)$, $H' = -iW$.

(a) Compute $k_\theta H k_\theta^{-1}$ and $k_\theta S k_\theta^{-1}$ as combinations of H and S.
(b) Deduce that $k_\theta E k_\theta^{-1} = e^{2i\theta}E$ and $k_\theta F k_\theta^{-1} = e^{-2i\theta}F$.
(c) Show that $k_\theta H' k_\theta^{-1} = H'$. List the K-types of $\mathrm{Ad}|_K$ on $\mathfrak{g}_{\mathbb{C}}$, with multiplicities.

Exercise 21.2. Let

$$H' = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad E = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad F = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}, \quad u_\pm = \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Let $V_2 = \mathrm{Sym}^2\mathbb{C}^2$, with basis $w_0 = u_-^2$, $w_1 = u_+u_-$, $w_2 = u_+^2$. A matrix X acts on products by the Leibniz rule $X(ab) = (Xa)b + a(Xb)$.

(a) Show that $H'u_\pm = \pm u_\pm$, $Eu_+ = 0$, $Eu_- = u_+$, $Fu_+ = u_-$ and $Fu_- = 0$.
(b) Write the matrices $D(H')$, $D(E)$, $D(F)$ of the action on V_2 in the basis (w_0, w_1, w_2). Check that $[D(E), D(F)] = D(H')$.
(c) Compute $D(\Omega)$ for $\Omega = H'^2 + 2EF + 2FE$.

Exercise 21.3. Let H', E, F be as in Exercise 21.2 and k_θ as in Exercise 21.1. Recall that $H = E+F$, $S = -i(E-F)$ and $W = iH'$ form a basis of $\mathfrak{g}$. Fix $\lambda \in \mathbb{C}$ and put $c_n = \tfrac{1}{4}\big(\lambda - (n-2)n\big)$ for $n \in 2\mathbb{Z}$. Let V_λ have basis $\{v_n : n \in 2\mathbb{Z}\}$. Define

$$k_\theta \cdot v_n = e^{in\theta}v_n, \qquad H' \cdot v_n = n v_n, \qquad E \cdot v_n = v_{n+2}, \qquad F \cdot v_n = c_n\, v_{n-2}.$$

Extend complex-linearly to $\mathfrak{g}_{\mathbb{C}}$, and let $\mathfrak{g}$ act by restriction.

(a) Check the relations $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$ on each v_n.
(b) Check the finiteness, derivative and covariance conditions (a)–(c) of a $(\mathfrak{g}, K)$-module. Show that V_λ is admissible.
(c) Show that $\Omega = H'^2 + 2H' + 4FE$ acts on V_λ by λ.
(d) Show that V_λ is irreducible if and only if $\lambda + 1$ is not the square of an odd integer.
(e) For $\lambda = 0$, find two nonzero proper submodules $U_2 \subset U_0$ and describe U_0/U_2.

Exercise 21.4. Let Π be a unitary representation of $\mathrm{SL}(2,\mathbb{R})$ on a Hilbert space $\mathcal{H}$. Let $V = \mathcal{H}_K^\infty$, and let $V(n)$ be its K-type n subspace. Let H', E, F be as in Exercise 21.2, and extend $d\Pi$ complex-linearly. Use the fact from Lecture 15 that $\langle d\Pi(X)u, w\rangle = -\langle u, d\Pi(X)w\rangle$ for $X \in \mathfrak{g}$ and $u, w \in \mathcal{H}^\infty$. Inner products are antilinear in the first slot. Assume $\Omega = H'^2 + 2EF + 2FE$ acts on V by a scalar λ.

(a) Show that $\langle d\Pi(Z)u, w\rangle = -\langle u, d\Pi(\overline{Z})w\rangle$ for $Z \in \mathfrak{g}_{\mathbb{C}}$, where the bar is conjugation with respect to $\mathfrak{g}$. Compute $\overline{E}$ and $\overline{H'}$.

(b) For $v \in V(n)$, show that

$$\|d\Pi(E)v\|^2 = \frac{n(n+2)-\lambda}{4}\|v\|^2, \qquad \|d\Pi(F)v\|^2 = \frac{n(n-2)-\lambda}{4}\|v\|^2.$$

(c) Suppose $V(0) \neq 0$. Show that $\lambda \leq 0$. Show also that the E-formula in (b), applied to a single even K-type n, never gives a bound stronger than $\lambda \leq 0$.

Exercise 21.5. Let $G = \mathbb{R}$ act on $L^2(\mathbb{R})$ by $(\Pi(t)f)(x) = f(x-t)$, and let $K = \{0\}$. Let f be the indicator function of $[0,1]$.

(a) Show that every vector of $L^2(\mathbb{R})$ is K-finite.

(b) Compute $\|\Pi(t)f - f\|^2$ for $|t| \leq 1$.

(c) Show that f is not a smooth vector.

Exercise 21.6. Let V_λ and V_μ be the modules of Exercise 21.3, with bases v_n and v'_n and coefficients $c_n(\lambda)$ and $c_n(\mu)$. Let $T : V_\lambda \to V_\mu$ be a morphism of $(\mathfrak{g}, K)$-modules.

(a) Show that $Tv_n = t_n v'_n$ for scalars t_n.

(b) Show that all t_n are equal. Show that $T \neq 0$ forces $\lambda = \mu$.

(c) Deduce that every endomorphism of V_λ is a scalar, even when V_λ is reducible.

Exercise 21.7. *Optional extension.* Let $G = \{g \in \mathrm{GL}(2,\mathbb{R}) : \det g = \pm 1\}$. Its Lie algebra is $\mathfrak{sl}(2,\mathbb{R})$, and $K = \mathrm{O}(2)$ is a compact subgroup. Let k_θ, H', E, F be as in Exercises 21.1 and 21.2, and

$$r = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in \mathrm{O}(2).$$

As a matrix, r equals H. Here it is used as an element of $\mathrm{O}(2)$.

(a) Show that $rk_\theta r^{-1} = k_{-\theta}$, $rEr^{-1} = F$ and $rH'r^{-1} = -H'$.

(b) Let V be a $(\mathfrak{g}, K)$-module. Let $V(n)$ be the set of v with $k_\theta \cdot v = e^{in\theta}v$ for all θ. Show that $r \cdot V(n) = V(-n)$.

(c) On $V = \mathbb{C}$ let $\mathfrak{g}$ act by 0. Give two $(\mathfrak{g}, K)$-module structures on V that are not isomorphic. Explain why the derivative condition (b) cannot distinguish them.